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Characteristics of memory associations: A consumer-based brand equity perspective

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Abstract

This research uses a memory network model to identify various association characteristics underlying consumer-based brand equity. An empirical study measures association characteristics, such as set size, valence, uniqueness, and origin, and examines differences between high and low equity brands on these measures. The results show that consumer association differences are consistent with external equity indicators and provide insights on strong and weak areas for each brand that could be used to strengthen the brand. The discussion addresses implications of the association patterns for managing brand equity, compares the association measures with other equity measurement approaches, and broadens the set of association concepts used in this research.

Keywords: Memory associations; Brand equity

1. Introduction

Brand equity has recently emerged as an important research area in marketing (e.g., Barwise, 1993; Shocker et al., 1994) and has been examined from practical (Landor's global survey, Owen, 1993), strategic (Aaker, 1991; Kapferer, 1992), and theoretical perspectives (Keller, 1993). One may trace distinct areas of development of this equity concept. The initial interest in brand equity was from the standpoint of brand valuation. For example, when Unilever or Philip Morris acquired other companies, they were interested in a financial valuation of the acquired brands. This interest is reflected in one approach used by companies and researchers which is to model a brand's financial value based on various factors (e.g., Simon and Sullivan, 1993). For example, Interbrand (a leading brand equity consultant) models brand equity based on leadership, stability, the market environment, internationality, ongoing trends, support, and legal protection (Ourusoff and Panchapakesan, 1993). The information provided by this equity measurement approach is potentially useful for accounting purposes since it provides a financial estimate of a brand's value.

Apart from this financial valuation purpose, marketing managers may be interested in how to leverage the equity of strong brands. A second area of development has focused on how mature brands can leverage their equity through brand extensions, and the impact of such extensions (e.g., Aaker and Keller, 1990; Broniarczyk and Alba, 1994; Dacin and Smith, 1994; Loken and John, 1993; Reddy et al., 1994;

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Smith and Park, 1992; Sunde and Brodie, 1993). These endeavors enrich our understanding of how and when strong mature brands can be extended, and are thus directly relevant to a brand manager.

A third area of development focuses on measuring the equity of established brands from a customer perspective. For example, Swait et al. (1993) propose an Equalization Price measure, Kamakura and Russell (1993) discuss a Brand Value measure, and Park and Srinivasan (1994) introduce a consumer preference-based measure. Further, industry approaches such as Landor's ImagePower (see Owen, 1993) adopt a similar perspective by focusing on consumer measures of brand equity. The common assumption underlying this third perspective is that because customer-based measures underlie brand equity, any desired improvements to brand strength may be wrought through an analysis of such customer measures. The following comment by Barwise (1993, p. 100) aptly summarizes this perspective: "Arguably, until customer-based brand equity has been convincingly operationalized and measured, it is unrealistic to expect researchers to measure how it is increased and decreased by marketing actions."

However, these measures do not always provide guidelines for marketing actions. For example, when P&G acquired Old Spice in 1991 this brand was losing momentum on key measures such as share, price premium, and attitudes. Research confirmed that consumers' associations about Old Spice were generally negative (e.g., "outdated", "boring"). The normal reaction by any brand manager may have been to drop this weak brand. However, the brand group's analysis also revealed that consumers associated Old Spice with a "sailor ditty", "mariner man", "tall ships", and "male/female relationships". Based on this analysis, a strategy for reinvigorating the brand was developed that focused on bringing each of these associations up-to-date. The success of the subsequent ad campaign ("Wind in your Sails") based on this strategy was directly attributed to the recognition that Old Spice possessed some equity in the form of consumer associations which should be leveraged (rather than creating a new brand). Similar case analyses (e.g., Oldsmobile, Burger King) suggest that association measures can provide diagnostics to the brand manager that traditional sales measures do not. It is important to note that sales and association measures should usually be strongly correlated. The association measures that we develop in our research supplement other consumer measures by also providing the bases for improvements to a brand's strength.

Such market share parities co-occurring with strong differences in other bases of brand equity has been demonstrated in previous research. For example, Park and Srinivasan (1994) show that Listerine although equal in market share to Scope mouthwash, has significantly lower levels of equity. Further, although managers of private labels and low price brands may increase sales through promotion-based mechanisms, this may not translate into higher equity since the promotions may lower the brand's value in the consumers' minds (e.g., McQueen et al., 1993). Hence, a reliance on sales measures to assess brand strength may actually be detrimental to the brand since the value of the brand resides in the minds of customers and not in the brand itself.

Why might association measures be particularly relevant for understanding mature brands? For emerging brands, the brand manager is dealing with a cleaner slate since there are not many associations in the consumer's mind. Consumers may have seen a single ad for the brand and formed a judgment based on the ad or trial. Hence, the manager's main task is to induce the formation of some main associations between the product, brand, and its attributes. On the other hand, consider a mature brand such as Pepsi. Consumers have been exposed to this brand over several years (e.g., a current Pepsi television commercial tracks a consumer's purchase and usage of Pepsi over several decades) and may have developed a multitude of associations about the brand. Which of these associations should the brand manager attempt to emphasize in marketing communications? Further, the associations may be traced to various advertising campaigns and usage occasions, some of which possibly left enduring impressions on the consumer. Should the brand manager emphasize associations formed via trial more than those that are indirectly experienced? Further, some of these experiences may have been positive and others negative leaving the consumer with associations that are mixed in valence. Should the brand manager attempt to heighten the positive associations or remove the negative associations? For all these reasons, delving deeper into consumers' association structures (as in the Old Spice example) may provide new information to the brand manager on the brand's weaknesses and strengths. In addition, these measures provide a platform for creative use of associations in marketing the brand.

In summary, the perspectives adopted in this paper are important for the following reasons. First, the emergence of the equity concept was prompted by a need to measure the value of a brand (e.g., Aaker, 1991; Farquhar, 1989; Kapferer, 1992). The next logical step would be to provide bases for managerial actions, i.e., the equity measures should be diagnostic on areas of improvement for the brand. Second, while actual purchase behaviors may be used as measures of equity, delving into consumer knowledge structures may provide information on a brand's potential that may not be captured by past behaviors (McQueen et al., 1993; Keller, 1993; Park and Srinivasan, 1994). Finally, for a mature brand, any single measure is unlikely to capture the full range of its equity (e.g., Old Spice, Pepsi). Hence, a multitude of measures that reflect several dimensions of equity would be appropriate.

We build on Keller's approach to consumer-based brand equity. Keller defines equity as "the differential effect that brand knowledge has on consumer response to the marketing of that brand" (Keller, 1993, p. 2). Keller provides numerous ideas linking consumers' memory and knowledge (Alba and Hutchinson, 1987; Alba et al., 1991) to brand equity. However, the breadth of these consumer-based measures has not been explored from an equity perspective. More specifically, the notion that various characteristics of brand associations in consumers' memory may be used to indicate a brand's strength needs to be demonstrated.

Hence, the goal of this research is to address the above issues by exploring equity from a consumer perspective. Section 2 describes various properties of consumers' brand associations and how they indicate brand strength. Section 3 presents results from a study that explores whether high- and low-equity brands differ on various association characteristics. Section 4 develops implications of the association patterns for brand equity, compares various approaches to measuring equity, and raises some issues for future research.

2. Background: Consumer associations

A memory model is used to explain the key concepts relating to consumer associations. While memory models differ in form and underlying assumptions (e.g., ACT, Anderson, 1983; MINERVA 2, Hintzman, 1986; TODAM, Murdock, 1982; SAM, Raaijmakers and Shiffrin, 1981; diffusion model, Ratcliff, 1978), many of them view memory in terms of links between various concepts. Memory is thus composed of knowledge that is organized as a network of connections (see Fig. 1). The building block of these models is a node, used to represent any piece of information (e.g., Nelson et al., 1993). For example, a node can represent a brand (Nike), a product (athletic shoes), or an attribute (durability). Links between any two nodes suggest an association in the consumer's mind. "Associations" then is used as a general term to represent a link between any two nodes. Other properties of associations such as strength and directionality are important (Anderson, 1983), but are not considered in this research.

Considerable evidence suggests that this network is a fuzzy structure that can take many forms based on the nature of the cues used to access this network (e.g., Barsalou, 1983). For example, when a consumer thinks of shoes, a particular set of associations is activated (e.g., running, exercise) whereas Nike may activate a different set of associations (e.g., swoosh, Michael Jordan). Since the current research focuses on brand equity, the spotlight will be on the set of associations that are activated in response to the brand name.

2.1. Number of associations

The number of associations evoked by a brand name is one variable that may be used to characterize equity. Over time, consumers build up an impressive set of associations about various brands. For example, the associations "athletic shoes", "durability", "swoosh", "Michael Jordan", "Greek Goddess" and "expensive" may be prompted by the brand name Nike. Some of these associations are brand attributes and benefits (e.g., durability) whereas others may represent each consumer's experiences with the brand. These idiosyncratic associations may be based on a multitude of sources over time, some

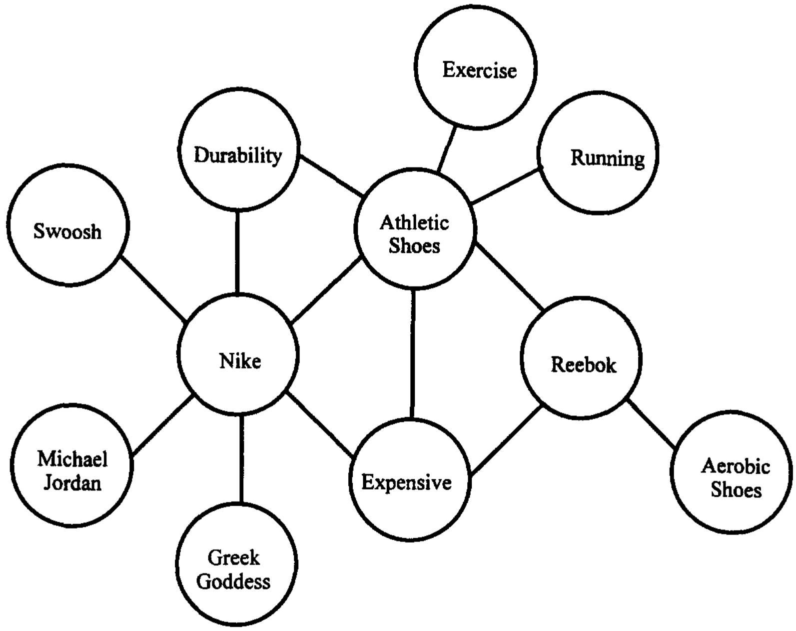

Fig. 1. Association network.

of which may be represented in memory as brand usage episodes. Many of these associations may not be product attribute-based, but it is valuable to understand these to provide a complete picture of how consumers perceive the brand. Park and Srinivasan (1994, p. 287) echo this sentiment, "Another avenue for further research that would assist brand managers involves developing a method that quantifies the importances of various brand associations unrelated to product attributes (e.g., the importance of the masculine image of Marlboro man) just as we measure the importance of product attributes in consumer preferences".

One might debate the desirability of a large number of brand associations. As the number of brand associations increases, the memory structure for that brand becomes richer, but also more complex. Typically, increasing the number of associations makes it easier to access the particular brand node from memory (e.g., associative network models) since these associations offer multiple pathways to the same brand node. On the other hand, a large number of associations may lead to lowered memory for the brand due to interference with these associations (e.g., Meyers-Levy, 1989); however, for mature brands (compared to new brands), this interference is not likely to be as strong since these brands have established higher levels of awareness (Kent and Allen, 1994). Hence, from an equity perspective, it is important for brands to have a large number of associations.

2.2. *Valence of associations*

Consumers store many brand associations in memory, some positive and others negative. Hence, focusing solely on the number of associations is misleading since many of these may be negative. For example, the brand name Chevrolet may prompt many associations such as "old", "cheap", "breaks down" which may all be considered negative by many consumers. Therefore, it is important to assess the relative presence of positive versus negative associations. Dacin and Smith (1994) indeed argue that "the favorability of consumers' predispositions toward a brand is perhaps the most basic of all brand associations and is at the core of many conceptualizations of brand strength/equity".

The advertising literature (e.g., MacKenzie and Lutz, 1989) suggests that one advertising objective may be to induce liking for the ad which is then expected to transfer to the brand (Mitchell and Olson, 1981). In fact this is such an important objective that some ad agencies, such as Leo Burnett, explicitly state this goal in their mission. In such cases, a common measure used to indicate the relative favorability of the ad is the net positive cognitive thoughts (number of positive minus number of negative responses). This measure can be adapted for the current research to reflect a focus on brands, and not advertising. For the Nike example provided earlier, the valence of each association could be assessed (athletic shoes, durability, swoosh, Michael Jordan, Greek Goddess are positive, whereas expensive is negative), and net valence can be computed on this basis. Note that in such cases, the total number of associations is controlled, with variations only in the net valence. Further, by focusing on all associations generated by the consumer, this process accounts for the impact of both product-attribute related and unrelated associations (Park and Srinivasan, 1994).

From a desirability standpoint, a strong brand should focus on consistently achieving net positive associations. Moreover, the strength of a brand may also be ascertained by considering how close these net positive associations are to ceiling levels. Brands that evoke mostly positive thoughts are likely to have net positive proportions close to 1 (Nike in Fig. 1). It is interesting to note that brands at the extremes in terms of equity (high or low) will have many associations. Only an examination of the valence of these associations provides a means of differentiating such brands.

2.3. Uniqueness of associations

Information about brands is part of a complex memory network that also includes information about the product category and other brands in the category. Some brand associations may be shared with the product, for example, "durability" may be shared by Nike and athletic shoes. The shared associations for Brand A are represented in Fig. 2 as A1, A2, A5, and A6, and the unique association is A7. The brand needs some shared associations with the category in order to be classified as a member of that category. Any mistakes in categorization may be detrimental to the brand; for example, when Sunlight detergent was first introduced, consumers mistakenly classified

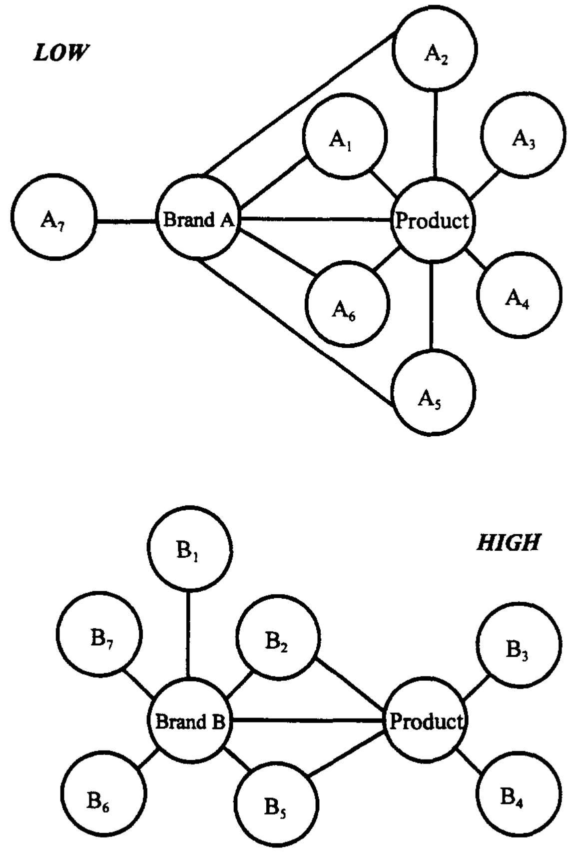

Fig. 2. Unique associations: from product.

the product as lemon juice due to a picture of a lemon on the new product's label.

As the number of shared associations increases, the brand increasingly becomes prototypical of the category, that is the brand increasingly becomes associated with standard product features. In such cases (e.g., Xerox, Post-It) the generic use of the brand name helps the brand by making it easier to recall and include in a purchase consideration set (Nedungadi and Hutchinson, 1985). However, this prototypicality can also be detrimental if consumers do not process new information/messages about the brand and instead rely on their prototypical impressions. Hence, the ideal situation for a high equity brand would be to have a large number of shared associations to be correctly and quickly classified as a member of that product category, but also possess some unique associations (e.g., swoosh for Nike) that enable it to stand out from the product category.

It is also conceivable that some associations are shared between brands, for example, "expensive" and "athletic shoes" may be shared between Nike and Reebok, whereas Nike possesses many unique associations such as "Greek Goddess", "Michael Jordan", and "swoosh" that Reebok does not possess. Unique associations are important since they contribute to the brand's image in the category (Broniarczyk and Alba, 1994; Farquhar, 1989; Keller, 1993) and are reflective of the brand's positioning in the consumer's mind. In fact, "unique selling proposition" has become a commonly used term in the brand manager's vocabulary when positioning a brand. Thus, the set of associations that are unique to a brand – relative to other brands in the category – may be used to indicate brand equity.

2.4. Origin of associations

Over time, consumers learn about products from a variety of sources. One reasonable position is that, for various reasons, associations from some sources are more important components of equity (Biel, 1993), since they lead to differences in association strength (Haugtvedt et al., 1993). One basic distinction in sources is between direct experience (e.g., trial, usage) and indirect experience with the brand (e.g., advertising, word-of-mouth). Compared with indirect experiences, associations based on direct brand experiences are likely to be more self-relevant (Burnkrant and Unnava, 1995), held with more certainty (Smith and Swinyard, 1983), and also form vivid autobiographical memories (Baumgartner et al., 1992). Hence, a brand that has a high proportion of associations based on direct experience should be in a relatively strong position, i.e., it possesses high equity. With indirect experiences, a further distinction may be made between non-marketer controlled (e.g., word-of-mouth) and marketer controlled (e.g., advertising) sources. From the consumer's perspective, higher degrees of credibility are attributed to the former source since there is no vested interest for word-of-mouth (e.g., Marks and Kamins, 1988). Thus, brands that have a large set of associations based on word-of-mouth benefit not only from the "free" communications but also from increased credibility; hence, presence of such associations may be viewed as equity indicators.

In summary, the association properties identified above form the basis of consumer-based brand equity. Measuring these association properties and comparing them across brands provides a means for exploring differences in consumer-based brand equity. The main expectation regarding the empirical results is that, even for brands that are well-known to consumers, differences in brand associations, and thus equity, can be demonstrated (e.g., Aaker, 1991). The following are the specific hypotheses for the study based on the discussion in this section.

Compared to brands with low equity, high equity brands will have:

H1: Greater number of associations;

H2: More net positive associations;

H3a: Fewer unique associations from the category;

H3b: More unique associations from competing brands;

H4a: More associations from direct experiences;

H4b: More associations from word-of-mouth.

Note that these measures are used as indicators, and not causes of consumer-based brand equity. From these equity indicators, a brand's weak and strong areas can be understood and the specific form of any ailments diagnosed. This is similar in spirit to a pathologist relying on a set of measures (e.g., temperature, blood pressure) to diagnose a patient's ailment. As in this analogy, the measures co-occur with the ailment and the specific pattern must be understood before a remedy can be prescribed. For example, for the Old Spice brand, poor sales results may be due to underinvestment in the brand which would be indicated by fewer number of associations that are positive and unique. On the other hand, Burger King in spite of huge advertising investments in the late 80s, experienced sales decline due to associations that were not positive or particularly relevant to the product category.

3. The study

Since the interest was in understanding equity, it was desirable to focus on brands in various categories that possess high levels of equity, as opposed to start-up brands. Moreover, since product category differences may exist, comparing brands within product categories was deemed necessary. Hence, the approach used in this study was to select two mature brands with known differences in equity (external index) from each of six product categories and measure the association properties for these brands. The validity of the associations approach to measuring brand equity can be assessed by (a) testing differences between high and low equity brands on each of the association characteristics, and (b) highlighting actions that the association patterns suggest for the low-equity brands.

3.1. Method

The brands used in the Landor survey (Owen, 1993) provided a useful starting point since this was a comprehensive global study that cut across product categories. From this published list, a subset of products was chosen that would be relevant to both males and females in the student sample used in this study. Further, these categories were chosen to reflect a diverse set of consumer products (food, alcoholic beverages, durables, personal care) to afford generalizability of the results. For each of these product categories, two strong mature brands were selected to examine equity differences. The Landor ranking or the US market share (for beer and shampoo) was used as an external index of brand strength/equity. Table 1 displays the Landor brand rankings (overall ranking, share-of-mind, and esteem) with the stronger brand (e.g., McDonald's) always listed first for each product category. [1] The last two rows of Table 1 also show the Landor rankings for two product categories that were used as controls for comparisons. The two brands in each of these categories possess similar equity levels, hence the association measures should not differentiate between these pairs of brands.

Study participants were recruited from a business school in the US and were offered course credit for their participation. An initial pretest was conducted with 35 students to assess levels of brand awareness and familiarity for the brands used in the study. Awareness was assessed through unaided recall by asking subjects to list as many brands as possible in various product categories; the proportion of subjects who listed the target brand was used to indicate awareness. Since subjects may not have remembered specific brands on this recall task, a familiarity task was used to assess whether they could recognize the brands. Familiarity was measured on a nine-point scale (not at all familiar – extremely familiar). Similar measures have been suggested by Keller (1993) and used in prior brand equity studies (e.g., Owen, 1993; Lane and Jacobson, 1995).

Table 1 shows the familiarity and awareness scores for brands used in the study. The results indicate that familiarity and awareness were extremely high for most brands, as expected. Pairwise comparisons between brands indicate that in three of the eight categories tested (pizza, television sets, shampoo), there were significant differences on the familiarity measures. The awareness results paralleled the above results, with the exception that two additional comparisons were significant (beer and fast foods category). These familiarity and awareness scores were, in general, consistent with the Landor external equity index. Interestingly, the brand comparisons showed differences between Sony and Sharp although they had equal shares, and no differences between Crest and Colgate although Crest had a higher share.

These pretest results also indicated that students were an appropriate sample for the products used in this study, since the familiarity scores were high. The subsequent data collection efforts focused on obtaining associations data for pairs of brands in the eight categories. Overall, a total of 240 subjects participated in the study. Each subject provided associations data for a total of four brands from two product categories, and the products themselves (to be used in computing uniqueness). This resulted in an average sample size of 30 for each measure reported in the subsequent sections.

3.2. Measures

Associations are measured in industry and consumer research in a number of ways. Aaker (1991) categorizes these as direct methods that scale various brand perceptions and indirect methods that infer

[1] Only rankings available from Owen (1993) are reported in this table. The US market shares for the brands in the study are also listed in Table 1.

Table 1
External equity index for brands [a]

External equity = High (H) or Low (L)	Market share in US (%)	Landor ranking on overall equity	Landor ranking on share of mind	Landor ranking on esteem	Familiarity (1–9 scale)	Awareness (0 – 1 proportion)
Fast foods						
H: McDonald's	31.0	9	2	84	8.92 (0.41)	1.00 (0) *
L: Wendy's	8.1	100 +	100 +	100 +	8.31 (1.68)	0.89 (0.32)
Pizza						
H: Pizza Hut	26.1	77	34	190	8.38 (1.18) *	0.89 (0.32) **
L: Little Caesar's	11.2	100 +	100 +	100 +	7.59 (1.74)	0.64 (0.49)
Beer						
H: Budweiser	20.7	100 +	100 +	100 +	8.32 (1.37)	1.00 (0) **
L: Miller	3.5	100 +	100 +	100 +	8.09 (1.51)	0.75 (0.44)
Athletic shoes						
H: Nike	30.0	73	81	68	8.63 (0.79)	1.00 (0)
L: Reebok	22.0	100 +	100 +	100 +	8.05 (1.43)	0.97 (0.17)
Television sets						
H: Sony	31.5	37	60	27	8.20 (1.38) **	0.92 (0.28) ***
L: Sharp	31.0	100 +	100 +	100 +	6.37 (2.62)	0.06 (0.23)
Shampoo						
H: Pert	9.9	100 +	100 +	100 +	7.48 (2.07) **	0.78 (0.42) ***
L: Prell	2.4	100 +	100 +	100 +	5.61 (2.54)	0.22 (0.42)
Cola						
– Coke	20.4	1	1	5	8.91 (0.33)	1.00 (0)
– Pepsi	17.8	4	4	11	8.68 (0.74)	1.00 (0)
Toothpaste						
– Crest	29.4	22	31	0.28	8.45 (1.18)	0.97 (0.17)
– Colgate	18.4	28	23	45	7.78 (1.79)	0.89 (0.32)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
[a] The first column lists the market share and the next three columns are rankings of the top brands from Owen (1993). Brands not listed in the top 100 are labeled as "100 + ". The last two columns are mean values from this study. Standard deviations are in parentheses. Cola and toothpaste products are used as controls for comparisons.

meanings based on consumer responses. While direct methods usually provide quantifiable summaries of brand associations, indirect methods help the researcher "understand what a brand means to people" (Aaker, 1991, p. 137). Since the current focus was on understanding brand associations from the consumer's perspective, the latter approach was used in this study.

Number of associations. A free association procedure was used to assess the number of associations – subjects were asked to write down whatever came to mind when they thought of the brand in question. Subjects were specifically asked to think about the brand name each time they wrote down an additional response. This procedure has been shown to reduce response chaining since subjects have to think about the brand name after generating each association (e.g., Nelson et al., 1993). This procedure restricts the association quantity and content considerably since other associations to the brand may be generated by using cues such as usage occasions, product category, etc. However, since the current focus was on comparing brands, this procedure necessarily focuses on the (narrow) set of associations elicited by the brand. The number of thoughts generated in response to the brand name provides an indication of the associations linked to each brand name (Nelson et al., 1993). Hence, this approach incorporates strengths of the direct method (quantifiable summary) and indirect methods (listing of associations from the consumer's perspective).

Association valence and origin. In past research, independent coders would attempt to interpret the meaning of the responses from free association tasks. However, since independent coders would not be able to judge the valence and origin of these associations from each subject's perspective, subjects were asked to code their own responses. At the end of the free association tasks, subjects were instructed to examine each of their brand and product associations and code them into various categories. Since this coding task is fairly tedious, subjects coded their own responses on either valence or origin, to avoid fatigue. For valence, subjects coded each association on whether (for them) it was a positive, negative, or neutral association; a composite measure of net valence was used to represent the proportion of positive minus negative associations. Origin of each association was coded by subjects as to whether it was predominantly formed through advertising, direct experience, or word-of-mouth (family and friends). Since some of the associations may have multiple origins, the focus on the predominant source enabled each association to be ascribed to only one category. Since the three origin categories are qualitatively different, a single composite measure was not used to represent origin of associations.

Uniqueness. Two other measures were developed that focused on uniqueness of associations, with respect to the product category and to the competing brand. Since these were objective measures, an independent coder was used rather than instructing the subjects to code their own responses. For each brand, the number of associations that was shared by the brand and product was first counted; by definition, the unique associations to the brand can be calculated as the complement of the shared associations (proportion unique = 1 − proportion shared). For example, in Fig. 2, Brand A and Brand B each have five associations; however, Brand A shares four (80%) with the product whereas Brand B shares two (40%) with the product. Similarly, for the measure of uniqueness from the competing brand, the associations of the two brands were compared; only those associations mentioned for the focal brand but not for the comparison brand were considered unique and used to calculate the unique proportion.

3.3. Results

In this section, the differences between brands on the association measures are examined; evaluations of patterns across association measures are reserved for the implications section. Since the hypotheses predicted differences on associations as a function of the brand equity factor (high versus low), the data was analyzed using an ANOVA model. Each of the association measures was examined separately for differences between high and low equity brands. *T*-test comparisons between brands within each product category were used to test for statistical differences. For each measure, the means are first compared across product categories to provide a description of the data. Second, differences between brands within a product category are examined and compared with the external index to validate the current

Table 2
Brand comparisons: number, net valence, and uniqueness of associations [a]

External equity index = High (H) or Low (L)	Number of associations	Net valence	Uniqueness: from product	Uniqueness: from brand
Fast foods				
H: McDonald's	9.94	0.52	0.75	0.92
	(2.93)	(0.16)	(0.20)	(0.30)
L: Wendy's	6.96	0.49	0.91	0.89
	(1.91) ***	(0.15)	(0.23) ***	(0.27)
Pizza				
H: Pizza Hut	7.17	0.57	0.86	0.94
	(3.16)	(0.16)	(0.14)	(0.48)
L: Little Caesar's	5.69	0.70	0.92	0.92
	(2.53) *	(0.14) ***	(0.12) **	(0.43)
Beer				
H: Budweiser	7.91	0.47	0.87	0.85
	(3.62)	(0.18)	(0.23)	(0.44)
L: Miller	7.23	0.32	0.88	0.87
	(3.19)	(0.17) ***	(0.22)	(0.44)
Athletic shoes				
H: Nike	8.89	0.65	0.82	0.85
	(3.90)	(0.20)	(0.30)	(0.17)
L: Reebok	5.80	0.66	0.86	0.77
	(2.81) ***	(0.27)	(0.23)	(0.17) *
Television sets				
H: Sony	7.65	0.54	0.94	0.83
	(3.83)	(0.17)	(0.26)	(0.20)
L: Sharp	4.70	0.36	0.94	0.72
	(2.68) **	(0.28) ***	(0.29)	(0.22) *
Shampoo				
H: Pert	4.89	0.52	0.79	0.74
	(2.87)	(0.26)	(0.20)	(0.51)
L: Prell	3.57	0.35	0.65	0.78
	(1.97) *	(0.20) ***	(0.22) ***	(0.48)
Cola				
– Coke	6.21	0.46	0.82	0.89
	(3.57)	(0.27)	(0.38)	(0.79)
– Pepsi	8.11	0.41	0.82	0.91
	(3.68) *	(0.22)	(0.35)	(0.58)
Toothpaste				
– Crest	5.81	0.57	0.72	0.70
	(2.40)	(0.27)	(0.40)	(0.35)
– Colgate	5.55	0.59	0.70	0.69
	(2.94)	(0.38)	(0.40)	(0.48)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
[a] The first column represents the mean number of brand associations, while the next three columns are mean proportions, to control for differences in number of associations. Standard deviations are in parentheses. Cola and toothpaste products are used as controls for comparisons.

approach. Finally, the diagnostics provided by each measure are explained with reference to actions that brand managers can take to improve their equity.

Number of associations. Table 2 shows the mean number of associations for each brand in the study. Given that all these are mature brands and subjects are highly familiar with them, it is not surprising that the number of associations is high. Personal care products such as toothpaste and shampoo seemed to have lower associations compared to other products. There were many differences in the number of associations between brands in the same category. In five out of the six test product categories, there were significant differences between brands on the pairwise comparisons. Further, all of these differences were consistent with the external index. Surprisingly, one of the control products, cola, showed differences on this association measure. In this category, Pepsi had a higher number of associations compared to Coke, whereas they are roughly equal on the Landor external equity index and on market shares. [2] Perhaps, the recent targeting of Pepsi to young adults (e.g., "Pepsi: The Choice of a New Generation") has made this brand more relevant to this segment.

These results generally support the notion that the number of associations may be used to indicate equity (H1). Moreover, some of the absolute differences were fairly large, which may be used as an additional indicator of equity. The main recommendation for the brands with lower association scores (e.g., Wendy's) would be to systematically increase this set by making their brand more visible and relevant to the target market.

Valence of associations. As explained earlier, valence is indicated by net favorability (positive minus negative) and is indicated as a proportion to account for differences in the number of associations. Table 2 shows the means for valence of associations for each brand. Both brands in the athletic shoe category had high net valence proportions, suggesting that consumers have positive images of these brands. Further, even after controlling for the number of associations, valence differences emerge between brands. On four out of six test product comparisons, there were significant differences between brands within the same category (H2). Three of these were consistent with the Landor external index, whereas for pizza, Pizza Hut had a higher ranking on the Landor equity index and higher market share, but a lower net valence proportion. The use of a strong regional brand such as Little Caesar's as a contrast suggests that a brand's equity may vary by region. This implies that a brand with generally high levels of equity in Europe is not guaranteed of such strength within specific European countries.

The study results are consistent with prior research (Dacin and Smith, 1994) that regards valence of brand associations as an important indicator of brand equity. For brands that are weak on valence, the superior brands provide a useful benchmark as a potential objective. Also, for the stronger brands, there is considerable room for improvement since none of the valence proportions were at ceiling levels. For example, Little Caesar's can attempt to further distance itself from Pizza Hut by increasing the valence number from 0.70. This can take the form of either forming new positive associations or converting some of the negative associations into positive or at least neutral associations.

Uniqueness of associations. Uniqueness is defined and measured with respect to (a) the product category and (b) the competing brand. For the first measure, all the brands had more unique than common associations (in Table 2, all these proportions were greater than 0.5). Further, the absolute proportions were very high (greater than 0.80 for 11 out of 16 brands). Of the eight products, toothpaste brands had consistently lower proportions on uniqueness (0.72 and 0.70), indicating that these brands tended to be very similar to the overall product category. Finally, for the six test products, three pairwise brand comparisons (within the category) were significant; for the brands with the smaller proportions (McDonald's, Pizza Hut, and Prell), this indicates that they have successfully made themselves more prototypical of the category than other brands (H3a). At the same time, they also possess sufficient unique associations to stand out from the category. Two of these comparisons (McDonald's and Pizza Hut) were consistent with the Landor external equity index and market shares. The third comparison is puzzling since Pert has consistently outperformed Prell on the

[2] On the Landor survey, these rankings were consistent even for the 18–29 age group.

other association measures. This is an interesting pattern that needs to be addressed by future research.

For the second measure of uniqueness, most of the brands had high proportions on uniqueness. This is not surprising since these are strong brands that have unique images; moreover, if uniqueness were measured with respect to all the brands in the category rather than just one referent brand, this proportion is likely to be much smaller. On pairwise brand comparisons, two of the six test products showed significant differences (H3b). This revealed an interesting asymmetry between the strong (Nike and Sony) and weak brands (Reebok and Sharp) in terms of their unique associations; for example, the percentage of associations unique to Nike was 0.85 whereas for Reebok it was 0.77. These differences are consistent with the Landor external index and suggest that a unique brand image may be viewed as one element of consumer-based equity. Hence, rather than emphasizing associations that other brands already possess, a brand should carefully cultivate a unique image that cannot be readily copied or imitated by other brands. Interestingly, these differences were not reflected in market share differences between Sony and Sharp, suggesting that each measure may reflect a different component of brand strength.

Origin of associations. Table 3 shows the proportion of total associations attributed to various sources.[3] Only two of the pairwise brand comparisons (within category) were significant; associations for Pizza Hut from advertising were substantially higher (0.61 versus 0.33) whereas associations for Little Caesar's from direct experience were higher (0.67 versus 0.37). Since Pizza Hut has a higher ranking on the Landor equity index, the first result is inconsistent with our hypothesis (H4a). As discussed earlier, the second result may be due to Little Caesar's strong local presence and perhaps indicates that this brand possesses high levels of equity on a regional basis.

[3] Subjects were also provided "other" as one option in classifying the origin of associations. This fourth category and rounding may result in the proportions not adding up to 1 for any given brand.

Table 3
Brand comparisons: origin of associations [a]

External equity index = High (H) or Low (L)	Advertising	Direct experience	Word-of-mouth
Fast foods			
H: McDonald's	0.46	0.50	0.05
	(0.27)	(0.27)	(0.09)
L: Wendy's	0.41	0.56	0.04
	(0.21)	(0.23)	(0.08)
Pizza			
H: Pizza Hut	0.61	0.37	0.01
	(0.36)	(0.39)	(0.06)
L: Little Caesar's	0.33	0.67	0.03
	(0.27) **	(0.28) **	(0.07)
Beer			
H: Budweiser	0.58	0.35	0.01
	(0.28)	(0.29)	(0.09)
L: Miller	0.63	0.29	0.07
	(0.28)	(0.26)	(0.17)
Athletic shoes			
H: Nike	0.65	0.27	0.03
	(0.21)	(0.21)	(0.09)
L: Reebok	0.67	0.23	0.05
	(0.25)	(0.25)	(0.11)
Television sets			
H: Sony	0.55	0.31	0.09
	(0.30)	(0.29)	(0.14)
L: Sharp	0.52	0.27	0.11
	(0.38)	(0.32)	(0.20)
Shampoo			
H: Pert	0.44	0.44	0.09
	(0.35)	(0.35)	(0.26)
L: Prell	0.37	0.38	0.13
	(0.45)	(0.44)	(0.29)
Cola			
– Coke	0.55	0.37	0.01
	(0.26)	(0.22)	(0.08)
– Pepsi	0.51	0.42	0.04
	(0.29)	(0.29)	(0.11)
Toothpaste			
– Crest	0.45	0.44	0.03
	(0.35)	(0.32)	(0.09)
– Colgate	0.43	0.53	0.02
	(0.29)	(0.37)	(0.05)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
[a] The numbers in the three columns are mean proportions, to control for differences in number of associations. Standard deviations are in parentheses. Cola and toothpaste products are used as controls for comparisons.

The within-category comparisons did not support H4b. However, some interesting across-product differences were revealed by the data patterns. For beer and athletic shoes, associations from advertising dominated direct experience and word-of-mouth associations; perhaps this reflects the high media budgets supporting these products and subjects' high levels of involvement with these products. For brands in these categories, it is imperative that they find other credible sources to augment their advertising dollars. Direct experience associations were higher only for the fast food category, indicating that this is an important source of learning in this category. Word-of-mouth association scores were at or near floor levels for all products; this provides an interesting comparison with findings from the information search literature that show high levels of emphasis on word-of-mouth (e.g., Murray, 1991; Punj and Staelin, 1983). It is possible that consumers rely on word-of-mouth closer to purchase activities, but this information either does not get stored in memory or attributed to this particular source. For television sets and shampoo products, the marginally higher proportions for word-of-mouth (compared to other products) suggested that this was an important additional source of information about these products. Further, shampoo products had a good balance of associations between advertising and direct experience which may be a good model for brands in other categories (e.g., television sets) to emulate.

4. Implications

The goal of this study was to examine whether various association characteristics underlie consumer-based brand equity by comparing and validating these association measures with an external equity index. The results are largely supportive of the basic premise of differences between brands with known equity discrepancies. Moreover, while brand association differences are generally consistent with the external index of equity, we have tried to resolve any observed inconsistencies. Further, in each case, the identification of a brand's weakness on a particular association measure (e.g., valence) provides specifics on corrective measures. Hence, the association-based measures used in this research show much promise not only as measures of consumer-based equity, but also as diagnostics for mature brands.

4.1. Managerial issues: Understanding the association patterns

Whereas the earlier analysis focused on comparing brands on individual measures, from a managerial perspective, it is also important to examine the entire pattern of memory associations (e.g., Krishnan and Chakravarti, 1993, 1995). First, for a brand like Sony that is on par with or dominates Sharp on every measure, it is clear that the brand has built equity on several properties. Other brands that fit this profile are Nike and McDonald's. The challenge for these brands would be to further build on this strength and pull away from competitors on some or all properties; for example, a specific goal for Sony may be to increase net valence from 0.54 and direct experience-based associations from 0.31.

Conversely, for a brand such as Sharp that has not built much consumer-based equity, the challenge is to prioritize the areas for improvement. Should Sharp attempt to make consumers more knowledgeable about the brand by increasing the number of associations? Or should they try to change some of the negative associations that consumers already have about the brand? The association properties leading to the biggest disparity may, by necessity, be the first targets for improvement.

In comparison to these three cases (fast foods, athletic shoes, and television sets), a different pattern is observable in other product categories. Brands have built equity on some properties, but not on others. For example, Pizza Hut has a greater number of associations but is lower on net valence compared to Little Caesar's. Similarly, Pert has a mixed pattern. The question in these cases is two-fold: (a) how should these association properties be weighted in comparing equity of these brands?; (b) relatedly, should these brands attempt to strengthen these weaker properties even at the expense of some of the stronger ones? These tradeoffs further suggest that there are multiple interactions that need to be recognized when evaluating the equity dimensions.

4.2. Measurement issues

Since the current approach focuses on consumer-based brand equity, it is desirable to recognize the tradeoffs between various approaches to brand equity measurement. Our approach is different from the Landor and Interbrand approaches that we introduced earlier as exemplars of brand equity measurement. Landor is also consumer focused but is simplistic in that it relies on just two measures, awareness and image. Furthermore, Landor assigns equal weight to these two components, which may not be warranted in many categories. This measurement simplicity may also be its strength since results are easy to interpret. Furthermore, their approach allows a comparison of brand strength across countries using this consistent measure. Hence, compared to Landor, our measures are more complex but likely to be more diagnostic since they peel the layers of consumers' memory associations to understand what underlies brand strength.

Interbrand uses a well-known and widely used valuing approach that first focuses on a financial assessment of a brand's earning power that is directly contributed by the brand name. This financial measure is subsequently adjusted by a brand strength index which is based on seven factors. The strength of this approach is that it provides a single monetary value that can be used for accounting purposes. For example, Nike and Budweiser were valued at 3,497 and 8,243 million dollars each, whereas Reebok and Miller were valued at 2,251 and 1,018 million dollars respectively (Ourusoff and Panchapakesan, 1993). Note that these assessments are entirely consistent with both our results and the Landor rankings as to what constitutes a strong brand. Further, two of the Interbrand categories used to measure brand strength, trend and support, are designed to be directly relevant to a marketing manager by showing whether marketing investments are linked to the brand's financial strength. These measures capture first-order effects, however these categories are too broad to tell the brand manager where to invest to improve the brand's performance. Our approach focuses on second-order effects that can complement this analysis by showing specifically where the brand manager can make an investment to improve brand strength. By understanding the association properties, a brand manager can maintain current bases of strength for high equity brands by making them more unique (e.g., Nike), strengthen weaker brands by increasing the number of associations (e.g., Prell shampoo), or reinvigorate "tired" brands by bringing old associations up-to-date and making them more positive (e.g., Old Spice).

Hence, each of these three approaches has its strengths and weaknesses, and the appropriate method must be chosen based on the specific purpose of the firm. Our approach is perhaps of greatest relevance to a brand manager interested in managing a brand's strength. First, it provides an indication of various components of (consumer-based) strength which need to be understood and managed. Second, while our approach does not provide a financial measure it does benchmark competing brands (and not generics) which may be useful for comparison purposes (Barwise, 1993). Finally, and most importantly, each measure is directly relevant to a brand manager since these provide bases for developing specific strategies to improve the brand's position. As Keller (1993) argues eloquently, "Perhaps a firm's most valuable asset for improving marketing productivity is the knowledge that has been created about the brand in consumers' minds from the firm's investment in previous marketing programs. Financial valuation issues have little relevance if no underlying value for the brand has been created or if managers do not know how to exploit that value by developing profitable brand strategies".

4.3. Research issues: Expanding the model of memory associations

The model of consumer-based brand equity may be expanded further. First, the model presented in this research has not considered the impact of various moderating or mediating variables, such as consumer involvement (e.g., Laurent and Kapferer, 1985). Involvement may play a role not only at the brand level (by leading to strong differences between the favored brand and other brands) but also potentially at the product level (by heightening associations across the board for all brands within a category). Measuring brand and product involvement and assessing their specific role as moderators or mediators would be a useful enhancement to the model. Second, the model can also be expanded by considering differences between consumer segments. It is possible that with heterogeneous preferences, equity varies across segments. In such a case, the associations would have to be measured and summarized within segment and a segment-specific measure computed. This is similar in spirit to Landor's ranking of brands for various age segments. Finally, the model can be expanded by including association measures that tap implicit memory, in contrast to the explicit memory focus of this research. Such measures of implicit memory should provide indications of brand associations in memory that cannot be articulated by consumers, but that yet impact on their behaviors (Krishnan and Chakravarti, 1993; Krishnan and Shapiro, 1996).

Secondary associations. The core brand association concepts can be further extended. Since a brand's associations are part of a memory network, it is also useful to examine the (secondary) associations of these primary brand associations. The reason is that, over time, some of these secondary associations (Keller, 1993) may transfer to the brand. These secondary associations may then be leveraged by the brand in its marketing programs. For example, for Nike, the primary brand association "Michael Jordan" is in turn associated with "basketball", "success", and the "Chicago Bulls". This may be represented in Fig. 3 as C1, C2, and C3 for Brand A. Nike may attempt to transfer these secondary associations to Nike shoes by leveraging the primary association. For example, the link with "basketball" can directly help brand awareness, the link with "success" could be viewed as a brand personality, and the link with "Chicago Bulls" can augment efforts at building a regional presence (e.g., Chicago's Nike Town retail store). Hence, these secondary associations, although they may or may not be currently linked to the brand, offer much potential in the long run. In fact, consultants recommend the use of specific celebrities with specific brands based on their analysis of the celebrity image; these are the secondary associations. At the same time, an examination of the secondary associations provides a means for deciding which of the primary associations the brand should emphasize at any point in time. For example, if the secondary associations of "Michael Jordan" are more promising than those of "Greek Goddess", Nike should attempt to emphasize its link to Michael Jordan. Hence, the proposition that a brand with many secondary associations has higher equity needs to be tested in future research.

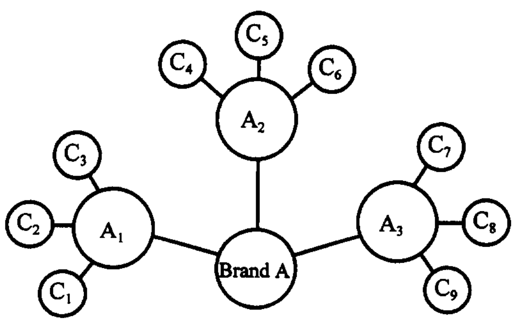

Fig. 3. Secondary associations.

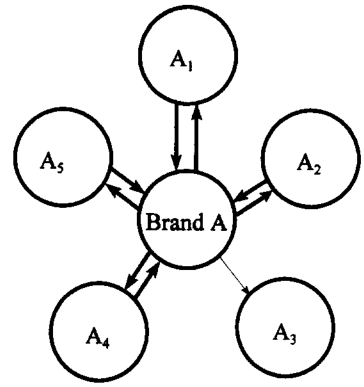

Fig. 4. Resonance of associations.

Resonance of associations. In general, brand associations focus on thoughts that come to mind when the brand is presented, hence a directionality from the brand to the association is assumed. If, when presented with the association, the brand comes to mind readily, the reverse arrow may also be drawn (see Fig. 4). This concept is known as resonance (Nelson et al., 1993). Consumers do not always think of the brand(s) first when they are making choices ("which brand has high *quality*" or "which brand is the best in this *product category*"). Hence, a high level of resonance would be desirable such that the brand is being cued by the various associations. For example, Nike would benefit by being cued any time consumers think of the swoosh, Michael Jordan, Greek Goddess, or athletic shoes (A1, A2, A4, and A5 in Fig. 4) since these associations resonate back to the brand. Thus, the proposition that high resonance may be viewed as an indicator of brand equity needs to be tested in the future.

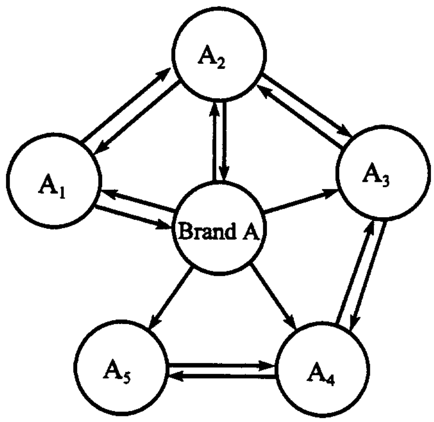

Fig. 5. Interconnectivity of associations.

Interconnectivity of associations. Fig. 5 provides a depiction of interconnectivity of associations, which may be employed to study potential third-order effects. As the word suggests, this is represented by a complex/simple network *among the primary associations*. For example, the links between A1–A2, A2–A3, A3–A4, and A4–A5 in Fig. 5 lead to a high level of interconnectivity for Brand A. High levels of interconnectivity enable these associations to be stored together, and retrieved as a whole since the various concepts cross-cue each other (Barsalou, 1983). The brand, located at the center of this network, benefits from this phenomenon. Therefore, the proposition that high levels of interconnectivity may be regarded as an equity indicator needs to be tested by future research.

In conclusion, this research builds upon various memory network models and Keller's brand equity framework to suggest a set of brand association concepts with direct relevance to consumer-based brand equity (Keller, 1993). The findings from the initial study are encouraging since they suggest differences between brands, within and across product categories. Future research needs to further build on this model for measuring and managing brand equity.

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